

DATA DICTIONARY

Database: HiddenGemMusic (SQL Server)

Project: Hidden Gem Music Discovery Platform | Course: SOFT290 / DATA206 | Student: Leena Komenski

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Dataset Overview

Source: Two Kaggle datasets: Spotify Historical Charts (2017–2021) and Top Spotify Songs in 73 Countries (2023–2025).

Coverage: Dataset 1: January 1, 2017 – December 28, 2021. Dataset 2: October 17, 2023 – ~March 2025.

Data Gap: 22-month gap between datasets (December 2021 – October 2023). All time-series visualizations mark this gap explicitly.

Total Rows: Approximately 28 million rows in FACT_ChartEntry combined.

Countries: 73+ countries. Dataset 1 uses full country names; Dataset 2 uses ISO 3166-1 alpha-2 codes. Bridged via the Country dimension table.

Granularity: One row per song, per country, per snapshot date (daily).

Schema: Star schema. FACT_ChartEntry is the central fact table surrounded by dimension tables. All API queries go through pre-computed summary tables only — ChartEntry is never queried at request time.

Viral 50 Note: Viral 50 entries (chart_type_id = 2) are excluded at the population stored procedure layer, not the read layer. This is because read SPs query pre-computed tables, not ChartEntry directly.

Summary Table Population Order

1. sp_PopulateSongCountryPresence
2. sp_PopulateSongCountryChart
3. sp_PopulateCountryYearStats
4. sp_PopulateGlobalOverlapByYear
5. sp_PopulateTrendVelocityBySong
6. sp_PopulateDiscoveryGapByDay
7. sp_PopulateIsolationScoreByCountry
8. sp_PopulatePeakReachBySong
9. sp_PopulateTopSongByCountryYear
10. sp_PopulateHiddenGems ← always last

Core Tables

DIM_Song [Dimension table]

One row per unique song. Central entity of the star schema. Audio feature columns (danceability through time_signature) are sourced from Dataset 2 only; they are NULL for songs exclusive to Dataset 1.

Column	Data Type	Description
song_id	int	Surrogate primary key, auto-incremented.
spotify_id	nvarchar(50)	Spotify track ID. Used as the cross-dataset matching key — parsed from the URL in Dataset 1, native in Dataset 2. NULL where parsing failed.
title	nvarchar(512)	Track title as it appears in the source dataset.

Column	Data Type	Description
duration_ms	int	Track duration in milliseconds. Dataset 2 only; NULL for Dataset 1-only songs.
is_explicit	bit	1 if the track is marked explicit; 0 if clean; NULL if not available. Dataset 2 only.
album_name	nvarchar(MAX)	Album or single name. nvarchar(MAX) to accommodate long titles with special characters. Dataset 2 only.
album_release_date	date	Album release date. Dataset 2 only; NULL if not available.
danceability	float	Spotify audio feature: 0.0–1.0 suitability for dancing based on tempo, rhythm, and beat strength. Dataset 2 only.
energy	float	Spotify audio feature: 0.0–1.0 measure of intensity and activity. Dataset 2 only.
key	tinyint	Spotify audio feature: musical key using pitch class notation (0=C, 1=C#/Db ... 11=B). Dataset 2 only.
loudness	float	Spotify audio feature: overall loudness in dB, typically –60 to 0. Dataset 2 only.
mode	tinyint	Spotify audio feature: 1 = major, 0 = minor. Dataset 2 only.
speechiness	float	Spotify audio feature: presence of spoken words. Values above 0.66 indicate mostly spoken content. Dataset 2 only.
acousticness	float	Spotify audio feature: 0.0–1.0 confidence measure of whether the track is acoustic. Dataset 2 only.
instrumentalness	float	Spotify audio feature: predicts whether a track has no vocals. Values above 0.5 indicate likely instrumental. Dataset 2 only.
liveness	float	Spotify audio feature: detects presence of a live audience. Values above 0.8 indicate live recording. Dataset 2 only.
valence	float	Spotify audio feature: 0.0–1.0 musical positiveness. High = happy/euphoric; low = sad/angry. Dataset 2 only.
tempo	float	Spotify audio feature: estimated beats per minute (BPM). Dataset 2 only.
time_signature	tinyint	Spotify audio feature: estimated beats per bar, typically 3–7. Dataset 2 only.

DIM_Artist *[Dimension table]*

One row per unique artist. Linked to songs through Bridge_SongArtist to support multi-artist tracks.

Column	Data Type	Description
artist_id	int	Surrogate primary key, auto-incremented.
artist_name	nvarchar(MAX)	Artist display name as it appears in the source dataset. nvarchar(MAX) to handle long names and Unicode characters. Uniqueness enforced at ingestion via NOT EXISTS logic — SQL Server cannot index MAX-length columns.

Bridge_SongArtist *[Bridge table]*

Resolves the many-to-many relationship between songs and artists. One row per song–artist pairing. artist_order = 1 identifies the primary/lead artist.

Column	Data Type	Description
song_id	int	Foreign key to DIM_Song. Part of composite primary key.
artist_id	int	Foreign key to DIM_Artist. Part of composite primary key.
artist_order	tinyint	Credit order on the track. 1 = primary/lead artist. Used for display ordering by stored procedures and the API layer.

Country *[Dimension table]*

One row per country. Geographic dimension used across all chart and summary tables. Populated from the instructor-provided CountryDataset.csv (245 countries), augmented with lat/long and region for globe rendering.

Column	Data Type	Description
country_id	int	Surrogate primary key, auto-incremented.
full_name	nvarchar(100)	Full English country name (e.g., 'United States').
iso_code	char(2)	ISO 3166-1 alpha-2 two-letter country code (e.g., 'US'). Cross-dataset join key and identifier for Mapbox globe rendering.
region	nvarchar(50)	Geographic region (e.g., 'Europe', 'Latin America'). Used for globe region filter. NULL for countries not assigned a region.
latitude	float	Geographic centroid latitude in decimal degrees. Used for globe dot placement in the frontend.
longitude	float	Geographic centroid longitude in decimal degrees. Used for globe dot placement in the frontend.

ChartType *[Lookup table]*

Lookup table defining the chart types present in the source datasets. All summary population stored procedures exclude Viral 50 entries (chart_type_id = 2) to prevent contamination of overlap and discovery gap metrics.

Column	Data Type	Description
chart_type_id	int	Surrogate primary key.
name	nvarchar(50)	Chart type name: 'Top 200', 'Viral 50', or 'Top 50'.
rank_start	smallint	Minimum rank in this chart type (always 1).
rank_end	smallint	Maximum rank in this chart type (200 for Top 200; 50 for Viral 50 and Top 50).

ChartEntry *[Fact table — ~28 million rows]*

Central fact table. One row per song–country–date combination. Contains every daily chart snapshot from both source datasets. This table is NEVER queried at request time — all application reads go through pre-computed summary tables. Viral 50 entries (chart_type_id = 2) are present but excluded at the population SP layer.

Column	Data Type	Description
chart_entry_id	int	Surrogate primary key, auto-incremented.
song_id	int	Foreign key to DIM_Song.
country_id	int	Foreign key to Country. NULL for Global chart entries from Dataset 1.

Column	Data Type	Description
chart_type_id	int	Foreign key to ChartType. 1 = Top 200, 2 = Viral 50. All summary SPs filter WHERE chart_type_id != 2.
snapshot_date	date	Date of this chart snapshot. Dataset 1 range: 2017-01-01 to 2021-12-28. Dataset 2 range: 2023-10-17 to ~2025-03-31. A 22-month gap exists between datasets.
rank	smallint	Chart rank on this date. 1 = highest charting.
streams	bigint	Raw stream count. Dataset 1 Top 200 only; NULL for all Dataset 2 entries.
popularity	tinyint	Spotify proprietary popularity score 0–100. Dataset 2 only; NULL for Dataset 1. Not comparable to stream counts.
daily_movement	smallint	Rank change from the previous day. NULL on a song's first appearance in a country. Dataset 1 only.
weekly_movement	smallint	Rank change from the same day the prior week. NULL if fewer than 7 days of prior data exist. Dataset 1 only.
trend	nvarchar(20)	Trend direction label from Dataset 1 source data (e.g., 'MOVE_UP', 'MOVE_DOWN', 'SAME_POSITION', 'NEW_ENTRY'). NULL for Dataset 2.
trend_velocity	decimal	Computed rank momentum score. Populated during the summary build pipeline. The canonical source for analytics is TrendVelocityBySong; treat this column as vestigial on ChartEntry.
source	nvarchar(50)	Hardcoded value: 'Dataset1' or 'Dataset2'. Identifies the originating dataset for each row.

Pre-Computed Summary Tables

All application API endpoints query these tables only. ChartEntry (~28M rows) is never touched at request time. Tables must be populated in dependency order — see the stored procedure documentation for the required execution sequence.

SongCountryPresence *[Summary — sp_PopulateSongCountryPresence]*

First table in the population dependency chain. Counts distinct countries each song charted in per year. Feeds CountryYearStats and GlobalOverlapByYear. Viral 50 entries excluded.

Column	Data Type	Description
presence_id	int	Surrogate primary key.
song_id	int	Foreign key to DIM_Song.
chart_year	int	Year extracted from snapshot_date.
country_count	int	Number of distinct countries this song charted in during chart_year.

TopSongByCountryYear *[Summary — sp_PopulateTopSongByCountryYear (always last)]*

One representative top song per country per year for lightweight globe card display. Used by the Discover page endpoint (sp_GetDiscoverPageInfo) to avoid scanning ChartEntry at request time.

Column	Data Type	Description
country_id	int	Foreign key to Country. Part of composite primary key with chart_year.
chart_year	int	Year. Part of composite primary key.
song_id	int	Foreign key to DIM_Song. The top-ranked song for this country and year.
album_name	nvarchar(512)	Denormalized album name for display without joins.
artist_name	nvarchar(MAX)	Denormalized primary artist name for display without joins.
song_name	nvarchar(512)	Denormalized song title for display without joins.

CountryYearStats *[Summary — sp_PopulateCountryYearStats]*

Aggregate chart statistics per country per year. Powers country profile views, KPI 1 (Global Overlap Rate), and overlap percentage display. Depends on SongCountryPresence.

Column	Data Type	Description
country_year_stat_id	int	Surrogate primary key.
country_id	int	Foreign key to Country.
chart_year	int	Year.
total_charted	int	Total distinct songs that charted in this country this year. Top 200 only.
shared_count	int	Songs this country shared with at least one other country in the same year.
unique_count	int	Songs exclusive to this country — did not chart anywhere else in the same year.
overlap_pct	decimal	$\text{shared_count} / \text{total_charted} \times 100$. Stored as a decimal (e.g., 74.32). Source metric for KPI 1.

GlobalOverlapByYear *[Summary — sp_PopulateGlobalOverlapByYear]*

Global-level overlap trend data aggregated by year. Powers the Global Overlap trend line chart on the dashboard. The is_gap flag marks the 22-month data gap so the frontend can render a visual break in the time series.

Column	Data Type	Description
overlap_year_id	int	Surrogate primary key.
period_label	nvarchar(10)	Human-readable period label (e.g., '2019').
period_year	int	Four-digit year.
period_month	int	Month number 1–12. NULL for annual-level rows.
overlap_pct	decimal	Global average overlap percentage across all countries for this period.
total_unique_songs	int	Count of distinct songs charting globally in this period.
songs_in_2plus	int	Count of songs that charted in 2 or more countries. Numerator for global overlap.
avg_countries	decimal	Average number of countries each charting song appeared in during this period.

Column	Data Type	Description
is_gap	bit	1 for the synthetic row representing the Dec 2021–Oct 2023 data gap. Signals the frontend to render a break or annotation in the trend line.

TrendVelocityBySong [Summary — sp_PopulateTrendVelocityBySong]

Measures how quickly a song gained chart presence across countries within a given year. Used in Hidden Gems scoring and the trend_score column on the HiddenGems table.

Column	Data Type	Description
velocity_id	int	Surrogate primary key.
song_id	int	Foreign key to DIM_Song.
country_id	int	Foreign key to Country.
chart_year	int	Year.
trend_velocity	decimal	Computed momentum score reflecting rate of rank improvement. Higher = faster rise. Cast explicitly as DECIMAL(10,4) during population to prevent NULL from implicit float conversion.

DiscoveryGapByDay [Summary — sp_PopulateDiscoveryGapByDay]

Measures days elapsed between a song's first chart appearance in any country (origin) and its first appearance in a second country (spread). Only entries with days_to_spread > 1 are included — values of 1 or less are excluded as likely simultaneous global rollouts. Cross-dataset entries can produce very large values; the median (~26 days) is the statistically defensible figure, not the mean (~144 days).

Column	Data Type	Description
discovery_gap_id	int	Surrogate primary key.
song_id	int	Foreign key to DIM_Song.
origin_country_id	int	Foreign key to Country. The country where the song first charted.
spread_country_id	int	Foreign key to Country. The country where the song subsequently appeared.
days_to_spread	int	Days between first chart date in origin and first chart date in spread country. Always > 1 (floor enforced at population).
bucket_label	nvarchar(10)	Spread speed bucket label. Values: '1-7d', '8-30d', '31-90d', '91-180d', '180d+'. The 1-7d bucket starts at day 2 due to the > 1 floor.
bucket_order	tinyint	Sort order for bucket_label (1–5). Ensures consistent chart ordering independent of label string sort.
first_chart_date	date	Absolute date of first chart appearance in origin_country_id.

IsolationScoreByCountry [Summary — sp_PopulateIsolationScoreByCountry]

Quantifies how musically isolated each country is relative to global charts. A high isolation score means the country's charts are dominated by locally unique songs not shared elsewhere. Feeds KPI 3 (Most Isolated Region).

Column	Data Type	Description
isolation_id	int	Surrogate primary key.
country_id	int	Foreign key to Country.
chart_year	int	Year.
local_songs	int	Songs exclusive to this country — not charting anywhere else this year.
shared_songs	int	Songs this country shared with at least one other country this year.
isolation_score	decimal	$\text{local_songs} / (\text{local_songs} + \text{shared_songs}) \times 100$. Higher = more isolated. Stored as a decimal percentage.
isolation_tier	nvarchar(10)	Categorical label derived from isolation_score. Values: 'Low', 'Medium', 'High'.
total_entries	int	Total chart entries for this country/year before deduplication. Used for minimum threshold filtering (≥ 100 entries required).

PeakReachBySong *[Summary — sp_PopulatePeakReachBySong]*

Records the single date on which each song reached its maximum simultaneous global presence (highest number of countries charting at the same time). Feeds KPI 4 (Peak Cross-Regional Reach).

Column	Data Type	Description
peak_reach_id	int	Surrogate primary key.
song_id	int	Foreign key to DIM_Song.
peak_date	date	Calendar date on which the song appeared in the most countries simultaneously.
country_count	int	Number of countries charting this song on peak_date.

HiddenGems *[Summary — sp_PopulateHiddenGems]*

Final output table. One row per country–song–year combination where a song is trending globally but absent from that country's charts. Must be populated last — depends on all other summary tables. The trend_score is sourced from TrendVelocityBySong.

Column	Data Type	Description
hidden_gem_id	int	Surrogate primary key.
country_id	int	Foreign key to Country. The target market where the song is absent.
song_id	int	Foreign key to DIM_Song. The globally trending song absent from country_id.
chart_year	int	Year of evaluation.
countries_charting	int	Number of countries where this song charted in chart_year. Higher = more globally established.
trend_score	decimal	Trend velocity score sourced from TrendVelocityBySong. NULL if no velocity data exists for this song/year.

SongCountryChart *[Summary — sp_PopulateSongCountryChart]*

Lightweight presence-check lookup. Tracks which songs charted in which countries in which years. Used for efficient membership checks in queries that need song/country/year presence without scanning ChartEntry. Composite primary key on (song_id, country_id, chart_year).

Column	Data Type	Description
song_id	int	Foreign key to DIM_Song. Part of composite primary key.
country_id	int	Foreign key to Country. Part of composite primary key.
chart_year	int	Year. Part of composite primary key.

Inactive / System Tables

Language [not in active use]

The Language table (language_id, language_name, language_code) was created during early schema design to support song language tagging. It has not been populated and is not referenced by any stored procedure or API endpoint. Retained in the schema for potential future use.

sysdiagrams [SQL Server system table]

Auto-created by SQL Server Management Studio for database diagram storage. Not part of the application schema.